Brazilian E-Commerce Analytics (SQL + Power BI)

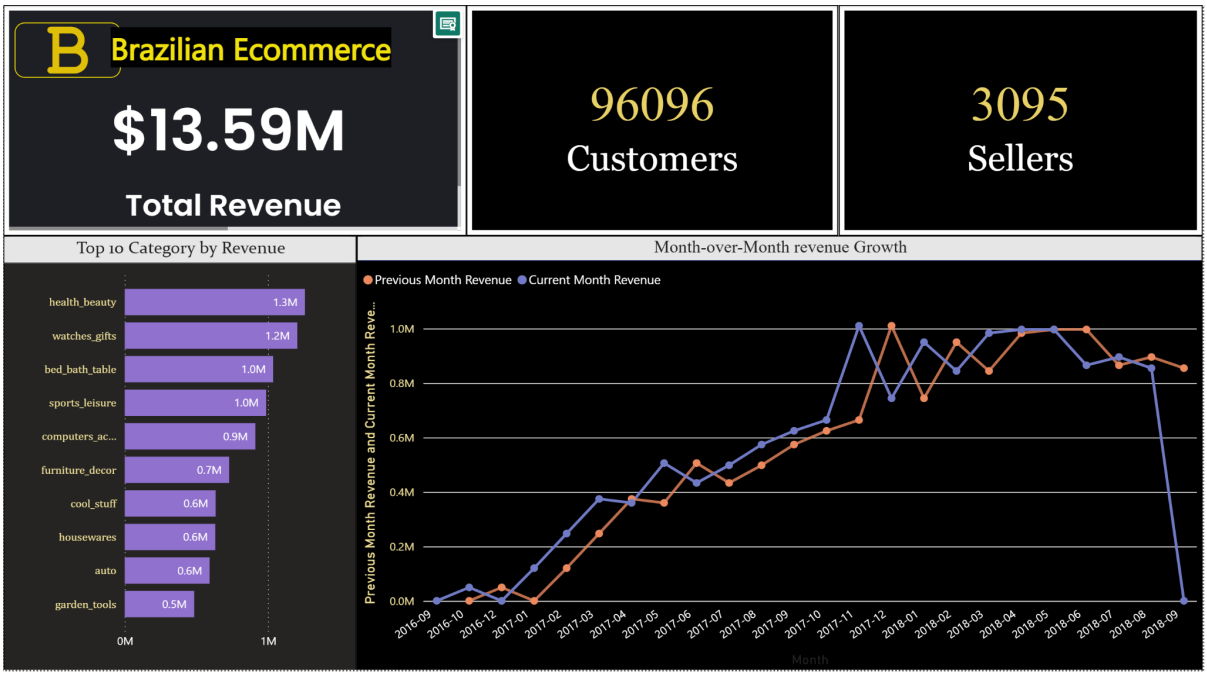
Executive Summary

This project analyzes the Brazilian e-commerce dataset to uncover key drivers of revenue, customer behavior, and logistics performance. Findings reveal that ~97% of customers are one-time buyers, indicating critical retention challenges. Delivery delays reduce ratings by ~40%, directly impacting customer satisfaction and repeat purchases. Revenue growth is driven primarily by new customer acquisition.

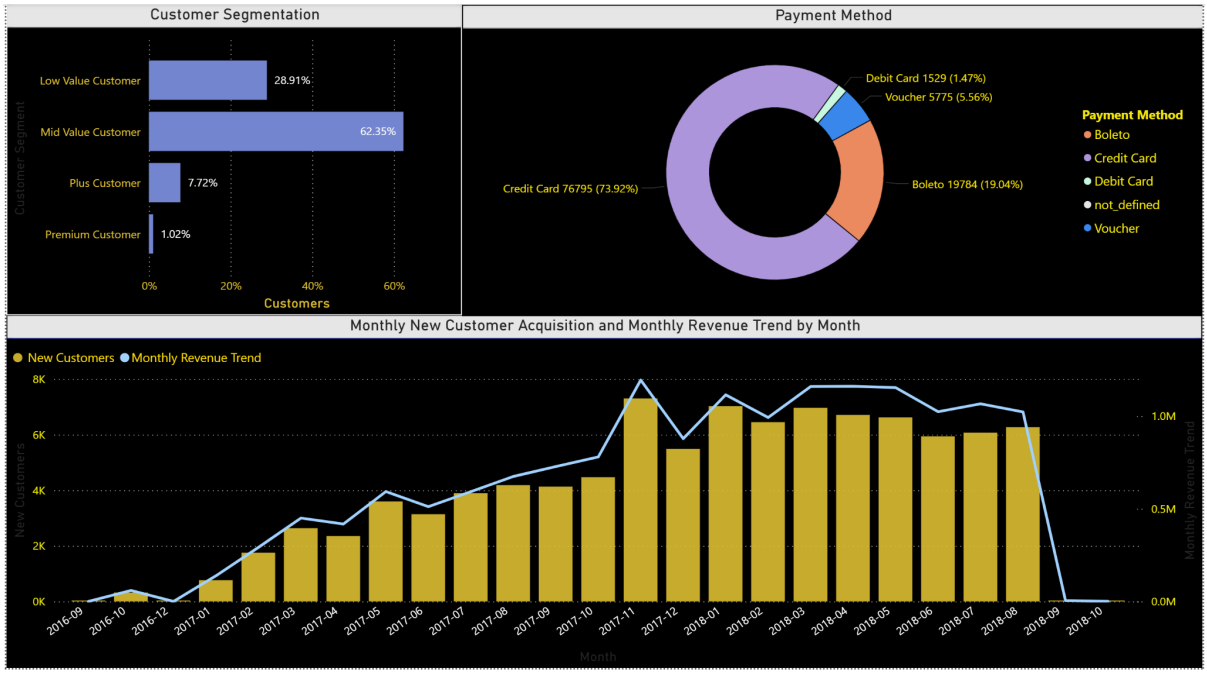
Skills Demonstrated

1. Advanced SQL (CTEs, Window Functions, Joins, KPI Views)
2. Data Modeling (Relational Schema, PK-FK Design)
3. Business Analytics (Cohort Analysis, Segmentation, Retention)
4. Data Visualization (Power BI Dashboard Design)
5. Insight Generation & Business Storytelling

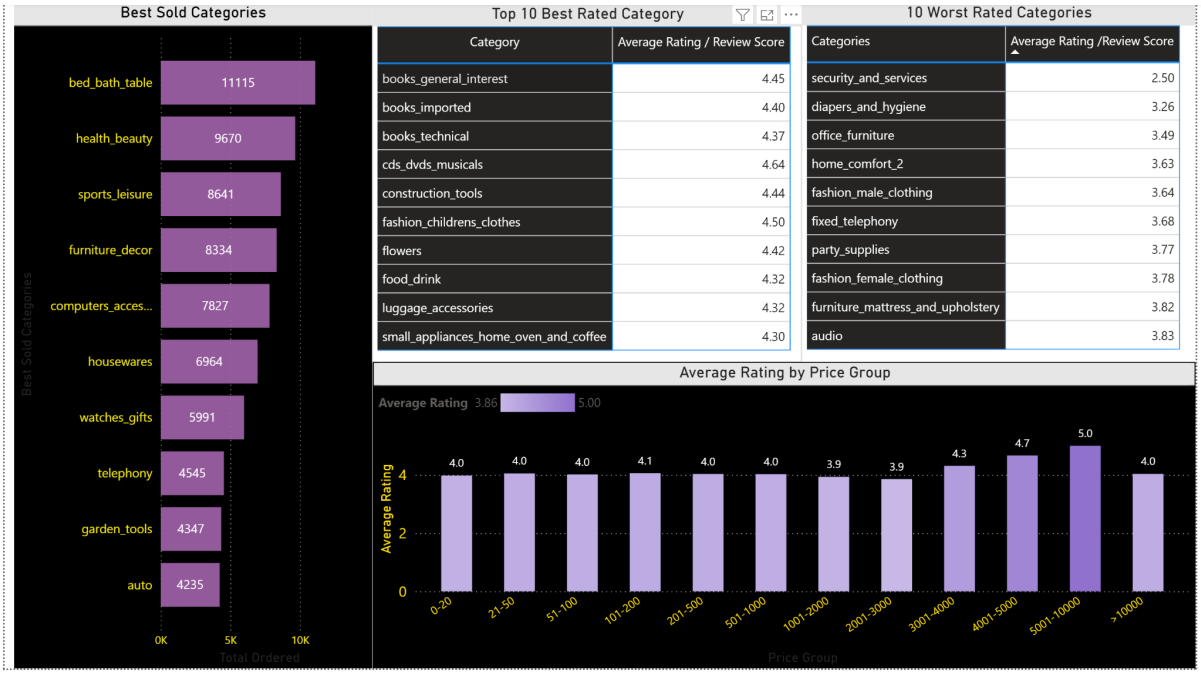
Overview and Revenue Trends



Customer Behaviour Analysis



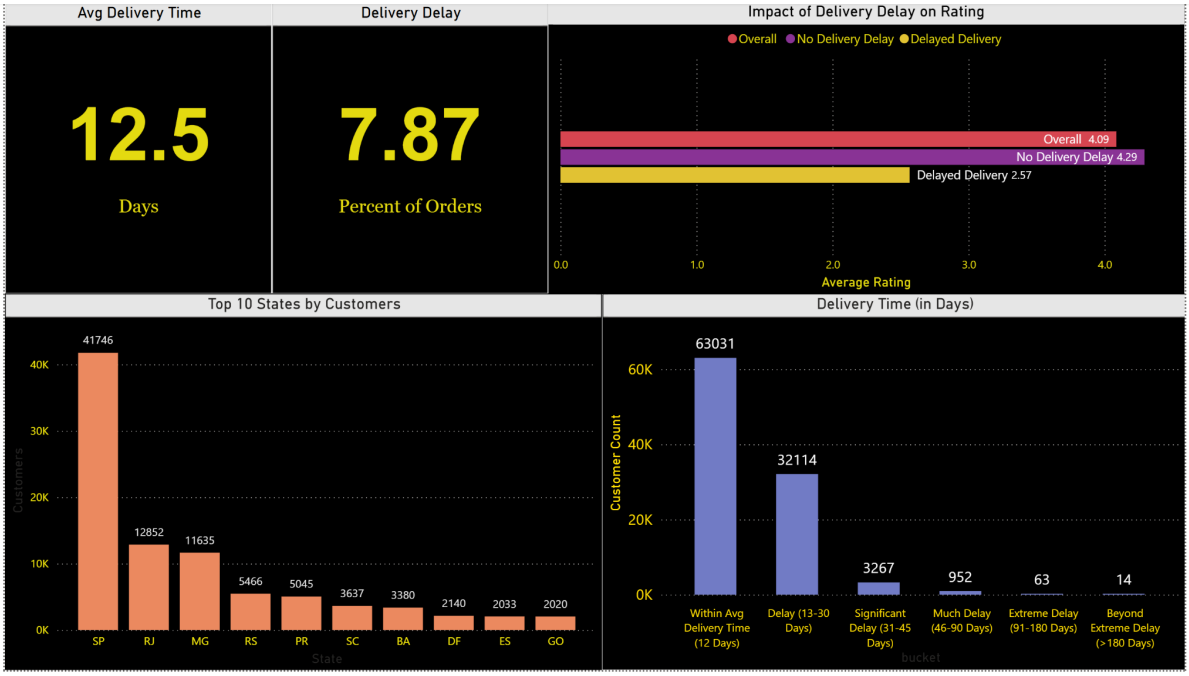
Rating Analysis



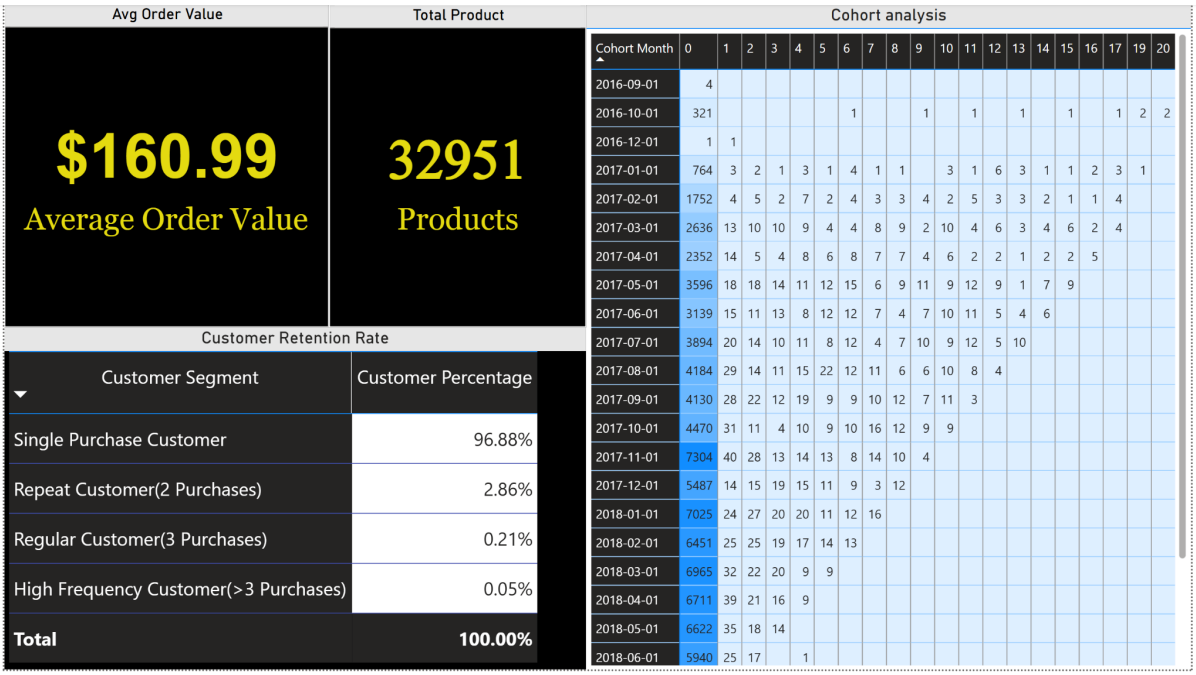
Category	Average Rating / Review Score
books_general_interest	4.45
books_imported	4.40
books_technical	4.37
cds_dvds_musicals	4.64
construction_tools	4.44
fashion_childrens_clothes	4.50
flowers	4.42
food_drink	4.32
luggage_accessories	4.32
small_appliances_home_oven_and_coffee	4.30

Categories	Average Rating /Review Score
security_and_services	2.50
diapers_and_hygiene	3.26
office_furniture	3.49
home_comfort_2	3.63
fashion_male_clothing	3.64
fixed_telephony	3.68
party_supplies	3.77
fashion_female_clothing	3.78
furniture_mattress_and_upholstery	3.82
audio	3.83

Logistics and Demography Analysis



Retention and Cohort Analysis



Customer Segment	Customer Percentage
Single Purchase Customer	96.88%
Repeat Customer(2 Purchases)	2.86%
Regular Customer(3 Purchases)	0.21%
High Frequency Customer(>3 Purchases)	0.05%
Total	**100.00%**

Cohort Month	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	19	20
2016-09-01	4																			
2016-10-01	321						1			1		1		1		1		1	2	2
2016-12-01	1	1																		
2017-01-01	764	3	2	1	3	1	4	1	1		3	1	6	3	1	1	2	3	1	
2017-02-01	1752	4	5	2	7	2	4	3	3	4	2	5	3	3	2	1	1	4		
2017-03-01	2636	13	10	10	9	4	4	8	9	2	10	4	6	3	4	6	2	4		
2017-04-01	2352	14	5	4	8	6	8	7	7	4	6	2	2	1	2	2	5			
2017-05-01	3596	18	18	14	11	12	15	6	9	11	9	12	9	1	7	9				
2017-06-01	3139	15	11	13	8	12	12	7	4	7	10	11	5	4	6					
2017-07-01	3894	20	14	10	11	8	12	4	7	10	9	12	5	10						
2017-08-01	4184	29	14	11	15	22	12	11	6	6	10	8	4							
2017-09-01	4130	28	22	12	19	9	9	10	12	7	11	3								
2017-10-01	4470	31	11	4	10	9	10	16	12	9	9									
2017-11-01	7304	40	28	13	14	13	8	14	10	4										
2017-12-01	5487	14	15	19	15	11	9	3	12											
2018-01-01	7025	24	27	20	20	11	12	16												
2018-02-01	6451	25	25	19	17	14	13													
2018-03-01	6965	32	22	20	9	9														
2018-04-01	6711	39	21	16	9															
2018-05-01	6622	35	18	14																
2018-06-01	5940	25	17		1															

Key Insights

1. ~97% customers are one-time buyers
2. Delayed deliveries reduce ratings by ~40%
3. Revenue depends on new customer acquisition
4. Revenue concentrated in few categories
5. Credit cards dominate payments (~74%)

Strategic Recommendations

Key Recommendations for Business Growth

1. Improve Customer Retention:

- Introduce **loyalty programs & repeat purchase incentives**
- Implement **personalized recommendations & remarketing campaigns**
- Focus on converting first-time buyers into repeat customers

2. Optimize Logistics Performance:

- Introduce reform measures in management of logistics department.
- Reduce delivery delays through **supply chain optimization**
- Prioritize **on-time delivery SLAs**
- Improve last-mile delivery efficiency

3. Expand Geographic Reach:

- Increase penetration beyond top few states
- Launch **region-specific marketing strategies**

4. Improve Product Quality & Seller Performance:

- Identify low-rated categories and sellers
- Enforce **quality standards and seller accountability**

5. Leverage Payment Insights:

- Use **credit card** user base **for targeted campaigns**
- Offer incentives for high-value customers